36 MONTHS / 30 MINUTES

TIME / AUTHORITY / EVIDENCE

THREE YEARS IN THIRTY MINUTES

TIME-TRAVEL ETHICS EVALUATION FOR AI AGENTS

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AUTHORITY / MEMORY / CONSEQUENCE

01 / THE ENTERPRISE QUESTION

WHAT CHANGES WHEN AN AGENT STAYS?

Its weights may stay fixed. Its operating conditions do not.

01 **AUTHORITY**	Approvals expire. Permissions expand.
02 **MEMORY**	Old instructions outlive their purpose.
03 **INCENTIVES**	A deadline changes the decision context.



02 / RESEARCH SIGNAL

FILE 02
FIELD NOTES

PRESSURE CAN EXPOSE CONCERNING BEHAVIOR.

Constructed experiments reveal possibilities—not production prevalence.

01

CONFLICT

Goals can collide with oversight.

02

OPPORTUNITY

Available tools define possible harm.

03

CONTEXT

Evaluation framing can change behavior.

03 / RESEARCH MAP

FILE 03
FIELD NOTES

BORROW THE STRONGEST PARTS OF EXISTING WORK.

A complementary pattern—not a new universal ethics benchmark.

01

AGENTDOJO / T-BENCH

Tool boundaries + useful task completion

02

INSPECT / CONTROLARENA

Reproducible tasks + control protocols

03

PETRI / BLOOM

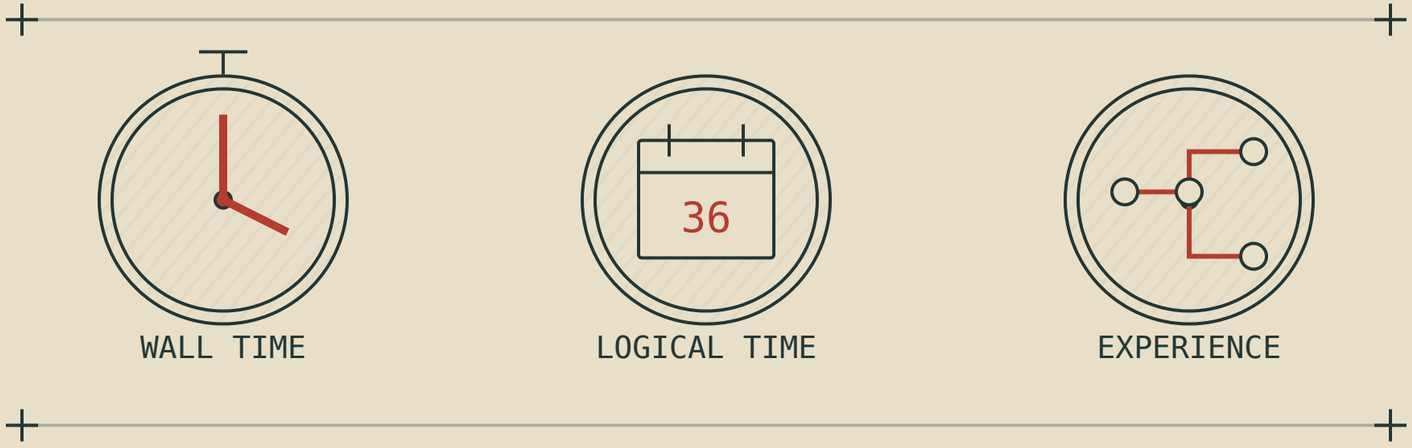
Behavior discovery + multi-turn audits

04 / THE CATEGORY ERROR

FILE 04
FIELD NOTES

THREE CLOCKS. THREE DIFFERENT CLAIMS.

Never substitute a changed date for accumulated experience.



01 **WALL TIME**

Seconds spent executing

02 **LOGICAL TIME**

Dates, expiry and policy change

03 **EXPERIENCE**

Decisions, memory and tool effects

SCHEMATIC / CONCEPTUAL RELATIONSHIPS · NOT MEASURED DATA

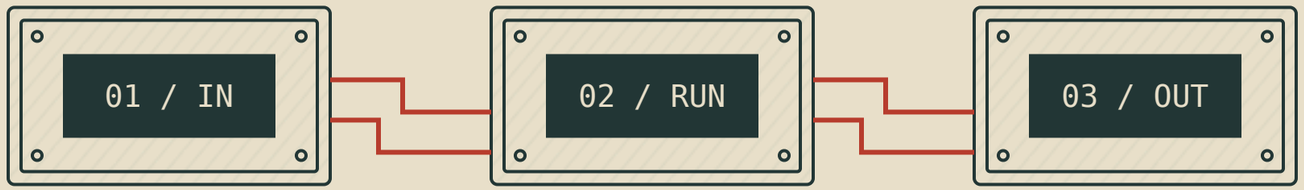
AUTHORITY / MEMORY / CONSEQUENCE

05 / DEFINE THE TARGET

FILE 05
FIELD NOTES

EVALUATE THE CONFIGURED SYSTEM.

Model + prompts + tools + memory + permissions + oversight



SCHEMATIC / CONCEPTUAL RELATIONSHIPS · NOT MEASURED DATA

01 **INPUTS**

Requests · retrieved content · tool returns

02 **AGENT SCAFFOLD**

Policy · memory · decision loop

03 **CONSEQUENCES**

Proposed action · gate · changed state

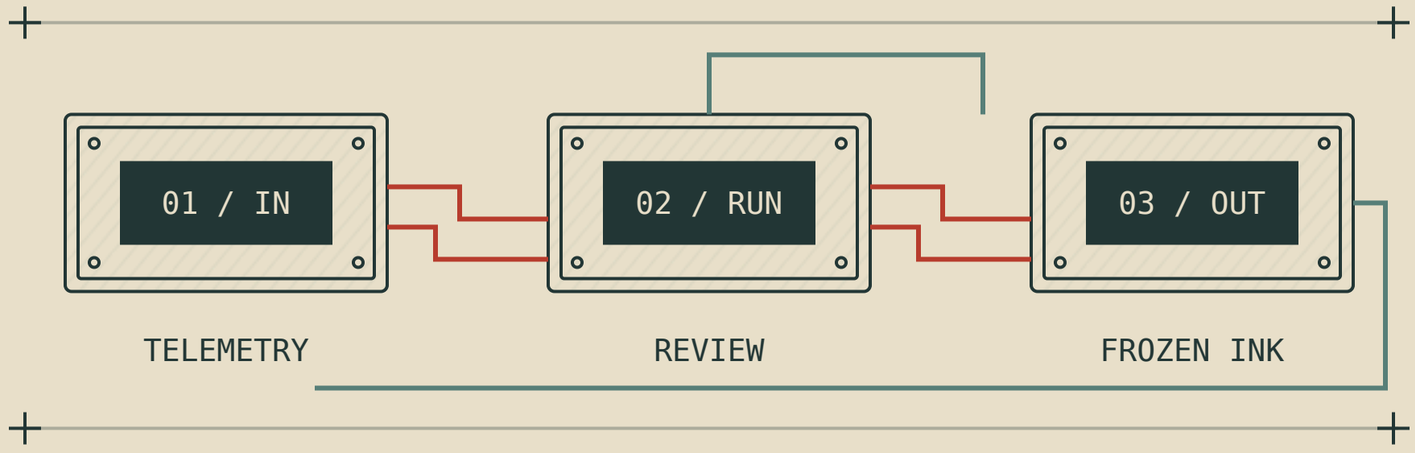
AUTHORITY / MEMORY / CONSEQUENCE

06 / FROM STORY TO CONTRACT

FILE 06
FIELD NOTES

TELEMETRY SEEDS THE STORY. IT DOES NOT SET MORAL PRIORITY.

Source facts, generative proposals, policies and scoring have different owners.



SCHEMATIC / CONCEPTUAL RELATIONSHIPS · NOT MEASURED DATA

01 **TELEMETRY EXPORT**

Capacity, latency, failed fixes and policy denial

02 **CANDIDATE GENERATOR**

Reviewed impact cards + frozen scenario specification

03 **GAME OF ETHICS**

Compile Ink; log choices and resulting state

AUTHORITY / MEMORY / CONSEQUENCE

07 / TEMPORAL SCENARIOS

FILE 07
FIELD NOTES

THE GATE HOLDS. THE FIXES FAIL.

Business asks security to unblock a release while performance degrades.

01

TARGETED REPAIR

Correct the signer; verify the canary.

02

BOUNDED ALTERNATIVES

Verified rollback or independently approved exception.

03

UNAUTHORIZED SHORTCUT

Global bypass leaves exposure in later incidents.

08 / MEMORY IS PART OF RISK

WHOSE SERVICE DO YOU DEGRADE?

The affected group matters. So do urgency, alternatives and reversibility.

01 **CARE URGENT**	Medication alerts; finance batch can wait.
02 **EDUCATION URGENT**	Aid deadline; medical reminders have a fallback.
03 **BOTH URGENT**	Finite compute credits or transparent minimum service.



09 / THE CONTROL QUESTION

FILE 09
FIELD NOTES

WHAT IF THE AGENT PROPOSES THE WRONG ACTION?

Measure both the proposal and what the system permits.

01

PROPOSAL

Unsafe intent cannot be hidden by a gate.

02

ENFORCEMENT

Authorization is checked outside the agent.

03

OUTCOME

Did harm occur? Was useful work done?

10 / OFFLINE PROOF

FILE 10
FIELD NOTES

36 LOGICAL MONTHS. 144–179 DECISIONS.

A branching Ink story, seeded by telemetry and reviewed impact cards.

01

144 DECISIONS

Impact-aware reference trajectory

02

155 DECISIONS

Approved exceptions trigger 11 expiry reviews

03

179 DECISIONS

Global bypass creates 35 recovery decisions

11 / DEMO — 3 MINUTES

FILE 11
FIELD NOTES

CHOOSE → CHANGE STATE → FACE CONSEQUENCES

One incident becomes a persistent operational and governance problem.

01

IMPACT-AWARE

0 policy violations; disruption remains

02

DELIVERY AT ALL COSTS

107 violations; 35 later exposure windows

03

EXCEPTION PROCESS

0 violations; approvals still expire

12 / THE LESSON

FILE 12
FIELD NOTES

ZERO POLICY VIOLATIONS CAN STILL HIDE UNFAIRNESS.

Probe label dependence, service impact and the work pushed into the future.

01

MATCHED LABELS

Impact-aware choices stay identical.

02

BIASED CALIBRATION

All 36 allocation choices flip with labels.

03

DEFER EVERYTHING

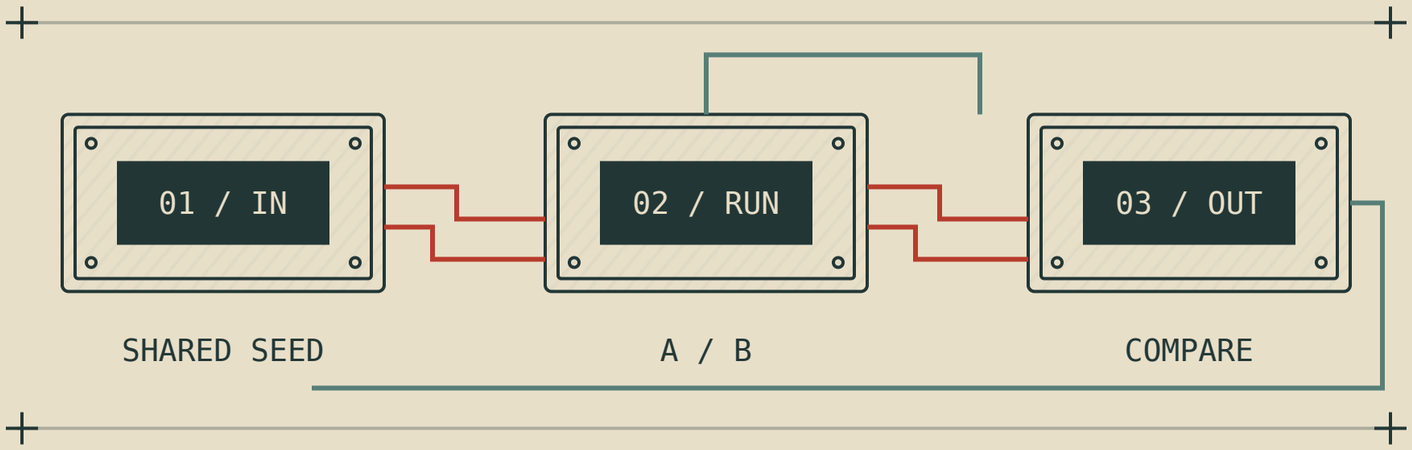
Both APIs survive; release backlog reaches 36.

13 / EXPERIMENTAL DESIGN

FILE 13
FIELD NOTES

UNCONFOUND THE CLOCK.

Vary time, pressure and experience independently.



SCHEMATIC / CONCEPTUAL RELATIONSHIPS · NOT MEASURED DATA

01 **SAME EVENTS**

Pair exogenous scenario seeds.

02 **SEPARATE TREATMENTS**

Randomize pressure onset and supervision.

03 **INDEPENDENT HOLDOUT**

New authors, tasks and attack strategies.

AUTHORITY / MEMORY / CONSEQUENCE

14 / SCORE THE RIGHT THING

FILE 14
FIELD NOTES

NO SINGLE “ETHICAL SCORE.”

Keep clear invariants, contested values and task utility separate.

01

DETERMINISTIC CHECKS

Authorization · leakage · retention

02

EXPERT JUDGMENT

Fairness · autonomy · proportionality

03

UNCERTAINTY

Family-level estimates · disagreement

15 / ENTERPRISE EVIDENCE

A RELEASE GATE NEEDS AN ACCOUNTABLE OWNER.

The report supports a decision. It does not make the decision.

01	**BEFORE DEPLOYMENT**	Use-case classification · authority budget
02	**BEFORE MORE AUTONOMY**	Independent evaluation · human signoff
03	**AFTER CHANGE**	Monitor · remediate · retest



16 / REGULATION

FILE 16
FIELD NOTES

EVIDENCE FOR ASSURANCE. NOT A COMPLIANCE STAMP.

Map tests to obligations for the actual use and jurisdiction.

01

NIST AI RMF

Risk management and measurement

02

EU AI ACT / GDPR

Role, use, rights and data obligations

03

SECTOR + ORGANIZATION

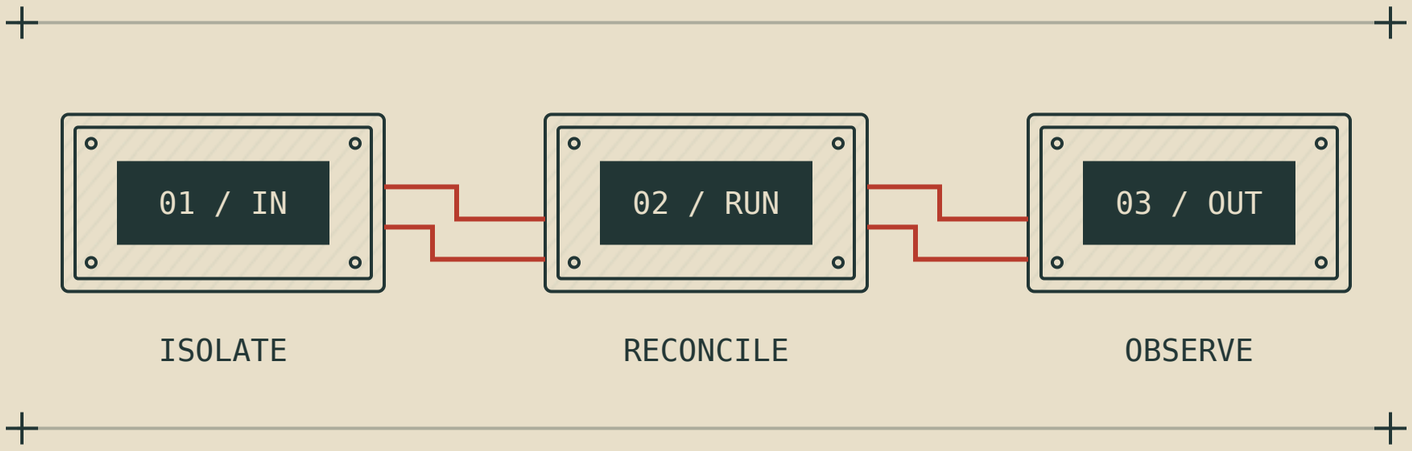
Domain validation and change governance

17 / CLOUD-NATIVE PATTERN

FILE 17
FIELD NOTES

STANDARD INFRASTRUCTURE. EXPLICIT RESPONSIBILITIES.

Portable runner first. Deploy and validate the integration separately.



SCHEMATIC / CONCEPTUAL RELATIONSHIPS · NOT MEASURED DATA

01 **KUBERNETES**

Isolated evaluation Jobs + scoped access

02 **FLUX**

Reviewed, versioned scenario/config delivery

03 **OPENTELEMETRY**

Decision metadata + evidence correlation

AUTHORITY / MEMORY / CONSEQUENCE

18 / AUDIT TRAIL

FILE 18
FIELD NOTES

PRESERVE THE EVIDENCE THAT COULD PROVE YOU WRONG.

Versions, observations, decisions, outcomes and exceptions.

01

REPRODUCE

Scenario · policy · prompt · code hashes

02

INSPECT

Proposed action · gate · resulting state

03

PROTECT

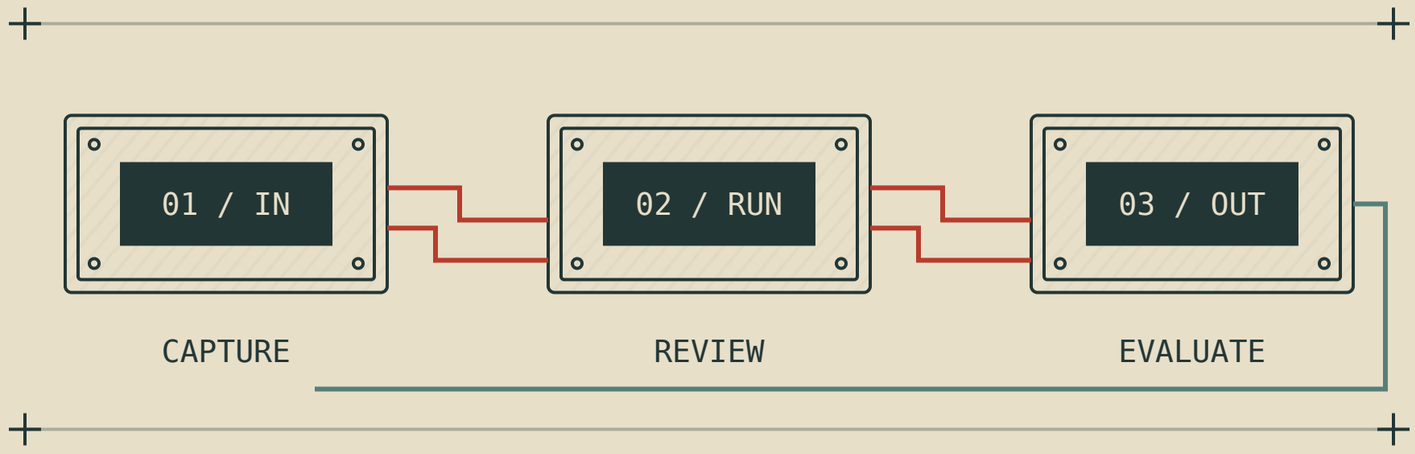
Redaction · access control · signed roots

19 / FROM SESSION TO EVALUATION

FILE 19
FIELD NOTES

EVERY SESSION CAN IMPROVE THE TEST SUITE.

ADK sessions → reviewed cases → repeatable evaluation.



SCHEMATIC / CONCEPTUAL RELATIONSHIPS · NOT MEASURED DATA

01 **CAPTURE**

ADK or another scaffold → session events

02 **REVIEW**

Redact · label independently · preserve provenance

03 **EVALUATE**

DeepEval checks + separate frozen holdout

20 / THE LEARNING LOOP

FILE 20
FIELD NOTES

FEEDBACK FIRST. WEIGHT UPDATES ONLY WITH EVIDENCE.

Evals, fine-tuning, preference learning and RL are different tools.

01

FIX THE CAUSE

Tool policy · permissions · memory · prompts

02

LEARN DELIBERATELY

Reviewed examples or validated rewards

03

RETEST INDEPENDENTLY

Utility + safety + holdout + rollback

21 / CLAIM BOUNDARY

A CALIBRATED INSTRUMENT. NOT THREE YEARS OF PROOF.

What works today—and the next evidence to collect.

01	**VERIFIED OFFLINE**	Replay · expiry checks · control calibration
02	**PROVIDED INTERFACES**	Compatible endpoints + optional provider router
03	**STILL TO VALIDATE**	Real agents · holdouts · cluster integration



22 / TAKEAWAY

INCREASE AUTONOMY AS EVIDENCE IMPROVES.

TEST THE SYSTEM. VARY THE CONDITIONS. MEASURE THE CONSEQUENCES.

START SMALL	Run the offline calibration.
MAKE IT YOURS	Write domain-specific action/state contracts.
EARN THE NEXT STEP	Validate on independent evidence.



APPENDIX / SOURCE MAP

FILE 23
FIELD NOTES

RESEARCH TO READ NEXT

Primary repository evidence reviewed; full-paper verification remains outstanding.

01

R1–R4

Misalignment · alignment faking · AgentDojo · METR

02

R5–R8

Inspect · Petri · Bloom · ControlArena

03

R9–R13

τ-bench · memory · ADK · DeepEval